Performance paradoxes in matching systems are not that rare

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Introduction

1. Introduction

2. Paradox in Star-free Graphs

3. (Multi)-Graph with self loops

4. Conclusion

Stochastic Matching model

Stochastic Matching model, Mairesse & Moyal, JAP, 2016

A Matching model is a triple (G, Φ, α) defined by a non-bipartite matching graph $G = (\mathcal{V}, \mathcal{E})$, a matching policy Φ and a distribution of probability α to model the arrivals of items.

- Items of the various classes in $\mathcal{V}$ arrive according to independent Poisson processes; their class i is drawn following α on $\mathcal{V}$.
- Any incoming item is matched, if possible, with a compatible item present in the system; Otherwise it is stored in a buffer.
- If several possible matches are possible, the incoming item follows a given matching policy Φ, such as First Come First matched (FCFM).

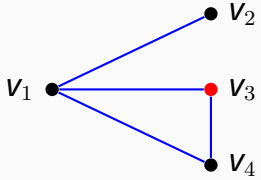
NCOND

The stability region of (G, Φ) is the set of probability measures α on $\mathcal{V}$ such that the natural Markov chain of (G, Φ, α) is positive recurrent.

NCOND

A necessary condition of stability under the FCFM discipline that $\alpha(\mathcal{I}) < \alpha(\Gamma(\mathcal{I}))$ for every independent set $\mathcal{I}$ in $\mathbb{I}$.

- $\mathcal{I} := \{i | \{i,j\} \notin \mathcal{E}\}$ for $\forall j \in \mathcal{I}\}$
- $\Gamma(\mathcal{I}) = \{j | \{i,j\} \in \mathcal{E}, i \in \mathcal{I}\}$



Expectation

$E[G, \alpha]$ is the expected total number of items in the system at steady-state.
For all independent sets $\mathcal{I} \in \mathbb{I}$, consider an ordered version of $\mathcal{I}$ noted $\mathcal{I}^o = \{i_1, \cdots, i_{|\mathcal{I}|}\}$, we note σ a permutation of its elements, i.e $\mathcal{I}^{\sigma(o)} = \{i_{\sigma(1)}, \cdots, i_{\sigma(|\mathcal{I}|)}\}$ and $\mathfrak{S}_{|\mathcal{I}|}$ the set of all permutations on the nodes of $\mathcal{I}$. We define:

$$T_{\mathcal{I}^o} = \prod_{k=1}^{|\mathcal{I}|} \frac{\alpha(\{i_k\})}{\alpha(\Gamma(\{i_1, \cdots, i_k\})) - \alpha(\{i_1, \cdots, i_k\})}.$$

Furthermore $T_{\mathcal{I}} = \sum_{\sigma \in \mathfrak{S}_{|\mathcal{I}|}} T_{\mathcal{I}^{\sigma(o)}}$. We also define:

$$E_{\mathcal{I}^o} = \sum_{l=1}^{|\mathcal{I}|} \frac{\alpha(\Gamma(\{i_1, \cdots, i_l\})}{\alpha(\Gamma(\{i_1, \cdots, i_l\})) - \alpha(\{i_1, \cdots, i_l\})} \prod_{k=1}^{|\mathcal{I}|} \frac{\alpha(\{i_k\})}{\alpha(\Gamma(\{i_1, \cdots, i_k\})) - \alpha(\{i_1, \cdots, i_k\})}$$

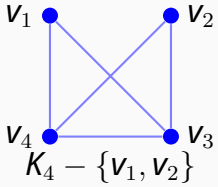
Finally, $E_{\mathcal{I}} = \sum_{\sigma \in \mathfrak{S}_{|\mathcal{I}|}} E_{\mathcal{I}^{\sigma(o)}}$. $\mathbb{E}[G, \alpha]$ can be rewritten as

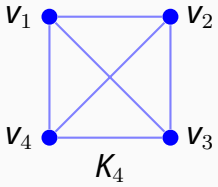
$$\mathbb{E}[G, \alpha] = \left(1 + \sum_{\mathcal{I} \in \mathbb{I}} T_{\mathcal{I}}\right)^{-1} \left(\sum_{\mathcal{I} \in \mathbb{I}} E_{\mathcal{I}}\right) \quad (1)$$

Performance Paradox

Cadas, Doncel, Fourneau & Busic. ACM SIGMETRICS, 2022

For some arrival rates α, we have a performance paradox if $\mathbb{E}[G + e, \alpha] > \mathbb{E}[G, \alpha]$.





EX.1	$\alpha = (0.21, 0.21, 0.475, 0.105)$	
	$\mathbb{E}[K_4 - \{v_1, v_2\}, \alpha] = 8.252$	$\mathbb{E}[K_4, \alpha] = 8.883$

Background

- A product form for the general stochastic matching model. Moyal, Busic, and Jean Mairesse. JAP, 2021
- Flexibility can hurt dynamic matching system performance. Cadas, Doncel, Fourneau and Busic. ACM SIGMETRICS, 2022
- A general stochastic matching model on multigraphs. Begeot, Marcovici, Moyal and Rahme. LAJPMS, 2021.
- $\mathbb{E}[G, \alpha]$ for the multigraphs ?

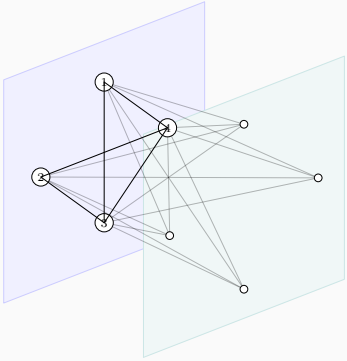
Property

Let $G1 = (V1, E1)$ and $G2 = (V2, E2)$ two arbitrary graphs, the join of $G1$ and $G2$ (denoted as $G1 \bowtie G2$) is defined by the set of nodes $V1 \cup V2$ and the set of edges $E1 \cup E2 \cup V1 \times V2$, where $\times$ is the Cartesian product.

Busic,Cadas,Doncel,Fourneau,Queueing Systems,2024

Performance paradoxes exist for graphs $IN_n \bowtie (K_4 - (1, 2))$ where IN_n is the graph with n isolated nodes.

The join of 4 isolated vertices and $K_4 - (1, 2)$: $IN_4 \bowtie (K_4 - (1, 2))$.

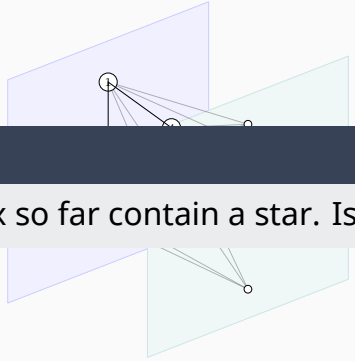


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Conjecture

All examples with paradox so far contain a star. Is it needed?

Paradox in Star-free Graphs

1. Introduction
2. Paradox in Star-free Graphs
3. (Multi)-Graph with self loops
4. Conclusion

The Minimal odd unicycle

EX.1	$\lambda = (0.111, 0.10575, 0.179426, 0.35046, 0.253364)$	
	$\mathbb{E}[C_5, \lambda] = 7.86$	$\mathbb{E}[C_5 + \{v_2, v_5\}, \lambda] = 8.17$
EX.2	$\lambda = (0.266025, 0.11005, 0.210855, 0.14145, 0.27162)$	
	$\mathbb{E}[F_1, \lambda] = 16.069$	$\mathbb{E}[F_1 + \{v_2, v_5\}, \lambda] = 16.89$

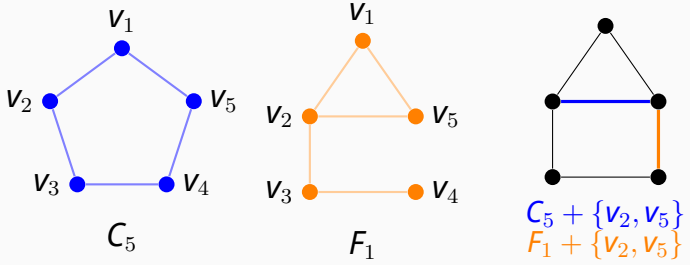


Figure: Graphs: C_5, F_1 and $C_5 + \{v_2, v_5\}$ (or $F_1 + \{v_2, v_5\}$).

The Minimal odd unicycle

EX.1	$\lambda = (0.111, 0.10575, 0.179426, 0.35046, 0.253364)$	
	$\mathbb{E}[C_5, \lambda] = 7.86$	$\mathbb{E}[C_5 + \{v_2, v_5\}, \lambda] = 8.17$

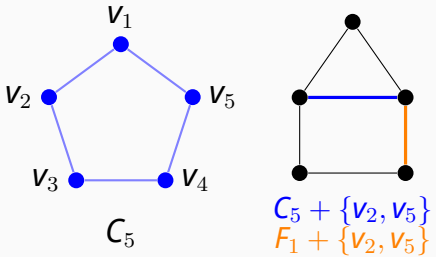


Figure: Graphs: C_5 and $C_5 + \{v_2, v_5\}$.

The Minimal odd unicycle

EX.2	$\lambda = (0.266025, 0.11005, 0.210855, 0.14145, 0.27162)$	
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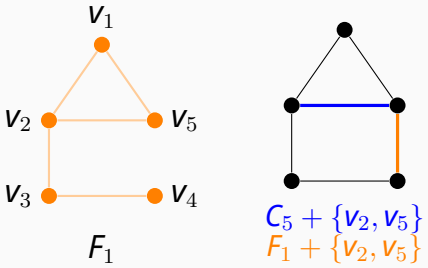


Figure: Graphs: F_1 and $F_1 + \{v_2, v_5\}$.

A Star-Free Graph with Two Odd Cycles

$\lambda = (0.231787, 0.462366, 0.14472, 0.127484, 0.00342, 0.030223)$	
$\mathbb{E}[F_3, \lambda] = 31.59$	$\mathbb{E}[F_3 + \{v_1, v_5\}, \lambda] = 31.71$

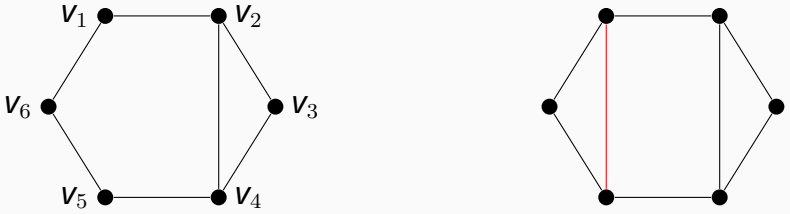


Figure: F_3 and $F_3 + \{v_1, v_5\}$.

C_7

$\lambda = (0.045206, 0.240384, 0.303789, 0.146114, 0.132526, 0.111772, 0.020209)$	
$\mathbb{E}[C_7, \lambda] = 129.56$	$\mathbb{E}[C_7 + \{v_1, v_5\}, \lambda] = 137.87$

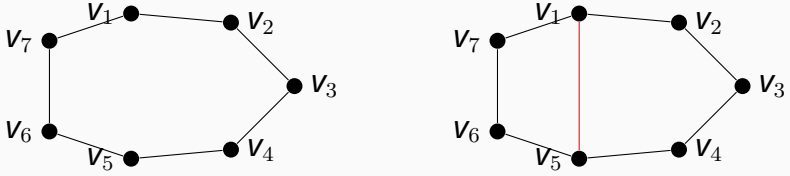


Figure: C_7 and $C_7 + \{v_1, v_5\}$.

(Multi)-Graph with self loops

1. Introduction

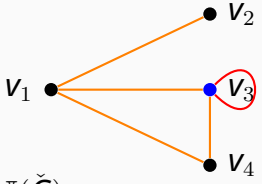
2. Paradox in Star-free Graphs

3. (Multi)-Graph with self loops

4. Conclusion

Definitions

- $G = (\mathcal{V}, \mathcal{E})$: a multigraph with the nodes partition $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$
- $\mathcal{V}_1 = \{i \in \mathcal{V} \mid \{i, i\} \in \mathcal{E}\}$ and $\mathcal{V}_2 = \mathcal{V} \setminus \mathcal{V}_1$
- $\check{G} = (\mathcal{V}, \mathcal{E} \setminus \{(i, i) \mid i \in \mathcal{V}_1\})$
- $\mathbb{I}(\check{G})$: the family of independent sets of $\check{G}$
- $\mathcal{I}^o = \{e_1, \ldots, e_{|\mathcal{I}|}\}$: an arbitrary ordering $\mathcal{I} \in \mathbb{I}(\check{G})$
- $\mathcal{I}^{\sigma(o)} = \{e_{\sigma(1)}, \ldots, e_{\sigma(|\mathcal{I}|)}\}$: the sequence obtained by applying σ to the ordered version $\mathcal{I}^o$
- $\mathcal{I}^o_k$: the first k elements of $\mathcal{I}^o$
- $\mathcal{J}^o_k = \mathcal{I}^o_k \cap \mathcal{V}_2$



Theorem

We define

$$T_{\mathcal{I}^o} = \prod_{k=1}^{|\mathcal{I}|} \frac{\alpha(e_k)}{\alpha\Big(\Gamma\big(\mathcal{I}_k^o\big)\Big) - \alpha\Big(\mathcal{J}_k^o\Big)},$$

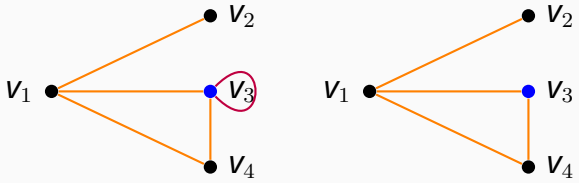
and $T_{\mathcal{I}} = \sum_{\sigma \in \mathfrak{S}_{|\mathcal{I}|}} T_{\mathcal{I}^{\sigma(o)}}$. Thus $\pi(\emptyset) = (1 + T_{\mathcal{I}})^{-1}$. We also define

$$E_{\mathcal{I}^o} = \sum_{l=1}^{|\mathcal{I}|} \frac{\alpha\Big(\Gamma\big(\mathcal{I}_l^o\big)\Big)}{\alpha\Big(\Gamma\big(\mathcal{I}_l^o\big)\Big) - \alpha\Big(\mathcal{J}_l^o\Big)} \prod_{k=1}^{|\mathcal{I}|} \frac{\alpha(e_k)}{\alpha\Big(\Gamma\big(\mathcal{I}_k^o\big)\Big) - \alpha\Big(\mathcal{J}_k^o\Big)},$$

and $E_{\mathcal{I}} = \sum_{\sigma \in \mathfrak{S}_{|\mathcal{I}|}} E_{\mathcal{I}^{\sigma(o)}}$. The expected number of items in steady state is

$$\mathbb{E}[G, \alpha] = \Big(1 + \sum_{\mathcal{I} \in \mathbb{I}(\check{G})} T_{\mathcal{I}}\Big)^{-1} \sum_{\mathcal{I} \in \mathbb{I}(\check{G})} E_{\mathcal{I}}. \tag{2}$$

Corollary



For simple graph.Cadas, Doncel, Fourneau,Busic. 2022

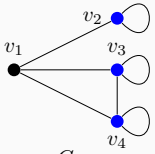
If $\mathcal{V}_1 = \emptyset$, then $\mathcal{J}_k^o = \mathcal{I}_k^o$.

Tools and Examples

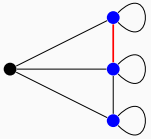




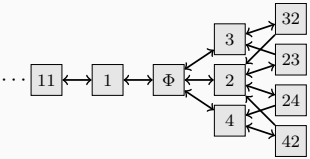


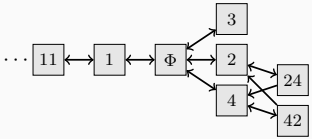


G



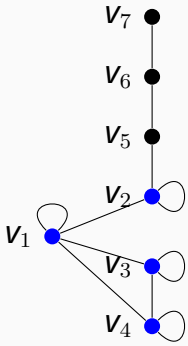
$G + \{v_2, v_3\}$



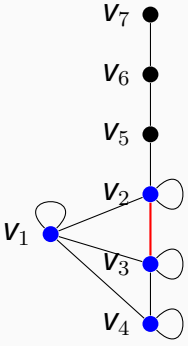


$\lambda = (0.432685, 0.005861, 0.141128, 0.420326)$	
$\mathbb{E}[G, \lambda] = 2.944793$	$\mathbb{E}[G_2, \lambda] = 2.947256$

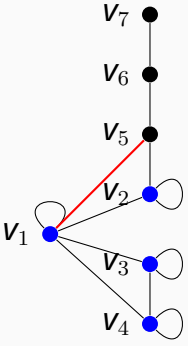
Examples



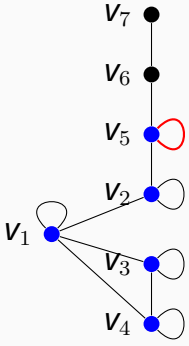
H



$H + \{v_2, v_3\}$

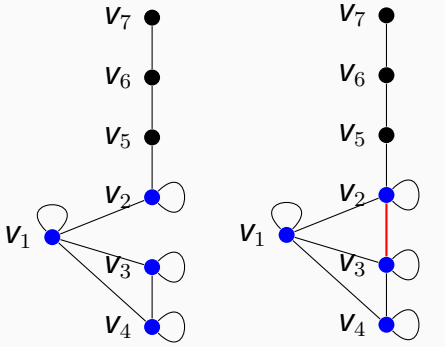


$H + \{v_1, v_5\}$



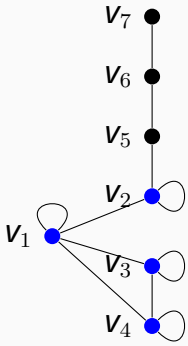
$H + \{v_5, v_5\}$

Examples

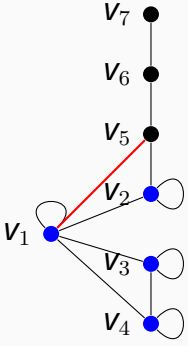


$\lambda = (0.070716, 0.180906, 0.238414, 0.204521, 0.164007, 0.07243, 0.069006)$	
$\mathbb{E}[H, \lambda] = 31.218542$	$\mathbb{E}[H + \{v_2, v_3\}, \lambda] = 31.539906$

Examples



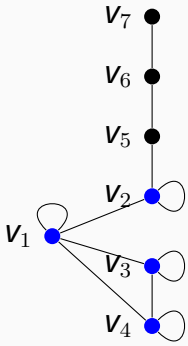
H



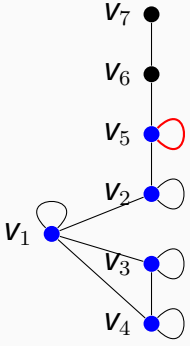
$H + \{v_1, v_5\}$

$\lambda = (0.08502, 0.06418, 0.076071, 0.096493, 0.336406, 0.326572, 0.015258)$	
$\mathbb{E}[H, \lambda] = 13.0179$	$\mathbb{E}[H_2, \lambda] = 13.058815$

Examples



H



$H + \{v_5, v_5\}$

$\lambda = (0.104627, 0.052725, 0.194586, 0.190934, 0.117413, 0.225466, 0.114249)$	
$\mathbb{E}[H, \lambda] = 33.838547$	$\mathbb{E}[H_3, \lambda] = 35.972599$

Conclusion

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Conclusion and Future work

- **Simple Graphs**
- Performed simulations for star-free graphs
- **Multiple Graphs**
- Derived the product-form expression for expectation
- Developed a program for simulation
- **Open Problems**
- No paradox under FCFM for graph with a loop for all node ?
- Explore other matching policies for possible paradoxical behavior

Questions?

Gracias Merci Thanks XieXie